

Zachary Nachod

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EDUCATION

University of Hawai‘i at Mānoa

Ph.D. Candidate in Marine Geology and Geochemistry

Concentrations: Chemical Oceanography, Data Science/Programming, Ocean Technology

Expected 2028

GPA: 4.0/4.0

University of Washington

Bachelor of Science in Oceanography

Minors: Arctic Studies, Applied Mathematics, Data Science

Graduated June 2022

GPA: 3.66/4.0

Field Experiences:

- A16S 2026 (March 9th – May 1st, 2026), R/V Revelle, 53 days
- HOT Cruise #356 (February 10th – 14th, 2025), R/V Sikuliaq, 5 days
- VISIONS Cruise 2021, R/V Thomspon, 10 days
- SMRU Consulting, Small Craft, Day Trips

RESEARCH EXPERIENCE

Research Interests: Autonomous Ocean Platforms and Global Observing Systems, Biogeochemistry, Data Quality Control, Southern Ocean and High-Latitude Oceanography, Ocean Mixing and Climate Change

Graduate Research Assistant

August 2022 – Present

University of Hawai‘i Department of Oceanography | Honolulu, Hawai‘i

Research Advisor: Seth Bushinsky

1. Developing a corrected float-based oxygen synthesis product that is consistent with shipboard measurements and internally consistent among BGC float observations.
 - Compared and combined multiple forms of oceanographic observations in Python.
 - Corrected data biases to create a data product that is accurate and internally consistent.
 - Performed data analysis and plotting in python showing impacts of biases and their removal.
2. Created an application for an outlier removal interface for Argo float data where the user can navigate through multiple floats and profiles to manually highlight, select, and remove data into an export format.
 - Utilized the dash and plotly packages in python to create an application on local host.
 - Worked directly with the lab, receiving input and feedback on various features of the application

Undergraduate Research Assistant

April 2019 – June 2022

University of Washington School of Oceanography | Seattle, Washington

Research Advisor: Alison Gray

1. Estimating biogeochemical properties on Seaglider tracks in the Southern Ocean.
 - Utilized MATLAB and Python to predict high resolution sea glider data from lower resolution SOCCOM float data and shipboard data using various machine learning models.
 - Implemented various statistical techniques to measure error in machine learning models.
 - Created pairwise plots as well as error histograms to adjust machine learning models for better accuracy.
2. Predicting movement of a Southern Ocean Carbon and Climate Observations and Modeling (SOCCOM) float in the Southern Ocean.
 - Analyzed satellite data paired with Argo floats to look at float movement reflected in sea surface height and predicted velocity.

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- Helped create a particle advection model based on sea surface height around the float, this model was then used to try and predict float movement.

Research Advisor: Susan Hautala

3. Investigating differences in Northern Pacific water column stratification in Coupled Model Intercomparison Project version 5 and 6 (CMIP 5/6) models.
 - Organized and manipulated large datasets in python using pandas and xarray.
 - Created various complex plots with the matplotlib and cartopy python packages.
 - Examined monthly averaged climatological fields from the various CMIP 5/6 simulations to better characterize differences in stratification and potential vorticity as well as to understand how these differences are related to large-scale ocean circulation.

CONFERENCE PRESENTATIONS

Nachod, Z. Detecting and Analyzing Bladder Problems in Argo Floats. Poster given at the University of Washington Undergraduate Research Symposium. Seattle, WA. (2021)

Nachod, Z., Gray, A.R. Evaluation of Machine Learning Techniques for Estimating Biogeochemical Properties on Seaglider Tracks in the Southern Ocean. Poster given at the University of Washington Undergraduate Research Symposium. Seattle, WA. (2022)

Nachod, Z., Bushinsky S.M., Tamsitt V., Williams N., Fassbender A.J. Development of an adjusted biogeochemical Argo float oxygen dataset to address observed biases in relation with shipboard data. Poster given at the Ocean Sciences Meeting. New Orleans, LA. (2024)

Nachod, Z., Bushinsky S.M., Tamsitt V., Williams N., Fassbender A.J. Development of an adjusted biogeochemical Argo float oxygen dataset to address observed biases in relation with shipboard data. Presented at Breakthroughs: Uehiro & Oceanography Yearly Symposium (BUOYS). Honolulu, HI. (2024)

Nachod, Z., Bushinsky S.M., Tamsitt V., Williams N., Fassbender A.J. Development of an adjusted biogeochemical Argo float oxygen dataset to address observed biases in relation with shipboard data. Poster given at the Southern Ocean Carbon and Climate Observations and Modeling Annual Meeting. Princeton, NJ. (2024)

Nachod, Z., Bushinsky S.M., Tamsitt V., Williams N., Fassbender A.J. Development of an adjusted biogeochemical Argo float oxygen dataset to address observed biases in relation with shipboard data. Presented at Breakthroughs: Uehiro & Oceanography Yearly Symposium (BUOYS). Honolulu, HI. (2025)

Nachod, Z., Bushinsky S.M., Tamsitt V., Williams N., Fassbender A.J. Development of an adjusted biogeochemical Argo float oxygen dataset to address observed biases in relation with shipboard data. Poster given at the Southern Ocean Carbon and Climate Observations and Modeling Annual Meeting. San Diego, CA. (2025)

Nachod, Z., Bushinsky S.M., Tamsitt V., Williams N., Fassbender A.J. Development of an adjusted biogeochemical Argo float oxygen dataset to address observed biases in relation with shipboard data. Poster given at the Ocean Sciences Meeting. Glasgow, Scotland. (2026)

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Nachod, Z., Bushinsky S.M., Jutras M., König D., McClish S., Addey C., Tamsitt V., Williams N., Fassbender A.J. BGC-Argo+: Global biogeochemical Argo float data with secondary quality control, derived parameters, outlier removal, and bias adjustment. Poster given at the Southern Ocean Carbon and Climate Observations and Modeling Annual Meeting. San Diego, CA. (2026)

PUBLICATIONS

Bushinsky, S. M., **Nachod, Z.**, Jutras†, M., König†, D., McClish*, S., & Addey*, C. BGC-Argo+: Global biogeochemical Argo float data with secondary quality control and derived parameters. Submitted.

Jutras, M., Bushinsky, S. M., Cerovečki, I., **Nachod, Z.**, & Hawco, N. Widespread increase of nitrate in subsurface waters over recent decades revealed by observations. ESS Open Archive, 2026(0213). <https://doi.org/10.22541/essoar.177100488.81138870/v1>

Bushinsky, S. M., **Nachod, Z.**, Fassbender, A. J., Tamsitt, V., Takeshita, Y., and Williams, N.: Offset Between Profiling Float and Shipboard Oxygen Observations at Depth Imparts Bias on Float pH and Derived pCO₂, Global Biogeochemical Cycles, 39, e2024GB008185, <https://doi.org/10.1029/2024GB008185>, 2025.

Sangmin Song, Paige D Lavin, Alison R Gray, **Zachary Nachod**, et al. Random forest regression using autonomous in situ ocean observations: Inferring small-scale nutrient variability during a Southern Ocean field experiment. ESS Open Archive. January 07, 2025. DOI: 10.22541/essoar.171707849.91867565/v2

WORK EXPERIENCE

Float Technician

July 2020 – August 2022

UW ARGO Float Lab | Seattle, Washington

Helps build, disassemble, test, analyze data, and ship/receive various APEX floats.

- Assembly, calibration, bench testing, and quality control of APEX, SOCCOM, and GO-BGC floats.
- Inventory and shipping management of oceanographic instruments.
- Data entry for the purpose of conducting quality control.
- Conduct data and failure analysis on large data sets to determine root cause with python.
- Experience training new hires in float assembly, maintenance, and data programming.

Research and Development Intern

June 2020 - September 2020

SMRU Consulting | Friday Harbor, Washington

Helped develop, build, and test numerous marine acoustic devices systems in contract with various customers such as the Maryland Department of Energy and the Port of Vancouver.

- Managed a project to redesign a mooring and the deployment/recovery plan of an acoustic device system.
- Worked in the field on small craft three days a week deploying and recovering coastal acoustic buoy systems.

Systems Engineer Intern

April 2018 – April 2020

Leidos | Gaithersburg, Maryland

Provided technical support and maintenance, either within the company or in support of customer contracts, on assignments using established procedures.

- Directly worked with customers to assess feedback for ongoing projects.
- Took on advanced responsibilities running project tests with the customer spanning weeks.

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- Utilized the Microsoft Office Suite to document, organize, and develop company information.

TEACHING EXPERIENCE

Lab Senior Mentor

September 2021 – August 2022

University of Washington | Seattle, Washington

Undergraduate Teaching Assistant

September 2019 – June 2022

University of Washington | Seattle, Washington

- OCEAN 101: Oceanography of the Pacific Northwest
- OCEAN 285/286: Physics Across Oceanography: Fluid Mechanics and Waves

Undergraduate Grader

March 2020 – June 2021

University of Washington | Seattle, Washington

- OCEAN 200: Introduction to Oceanography

HONORS & REWARDS

- Uehiro Center for the Advancement of Oceanography Fellow, 2025 - Present
- School of Oceanography Undergraduate Scholarship - 2021-2022 Academic Year
- School of Oceanography Lowell K. and Alice M. Barger Scholarship - 2020-2021 Academic Year
- Hugo Winkenwerder Fund - 2019-2020 Academic Year
- R.D. Merrill Forestry Scholarship - 2019-2020 Academic Year
- Agnes Healy Anderson Forestry Scholarship- 2019-2020 Academic Year
- Dean's List - Winter 2019, Spring 2019, Winter 2020, Autumn 2020, Winter 2021, Spring 2021

EXTRACURRICULAR

Current Activities:

- Nā Kama Kai - Vice President 2026-2027
- SOEST IDEAL 2025-Present
- Nā Kama Kai – Marine Geology and Geochemistry Representative 2025-2026
- Nā Kama Kai - Web manager 2023-24, 2024-25
- Nā Kama Kai - New Student Welcome Committee 2022-23

Previous Activities:

- Student Oceanographic Society - Vice President of Community Development and Diversity 2021-2022
- College of the Environment Grievance Committee - Undergraduate Oceanography Representative, Member 2022
- Washington Wave Team – Member 2021-2022
- UW ROV Team – Member 2021-2022
- UW Seed – Member 2018-2022
- UW Ecoreps – Member 2019-2022

Outreach Activities/Community Involvement:

- Hosted Lab Tours for students from Maryknoll High School April 23rd, 2025
- Volunteer for SOEST – Earth Fest at Kaiser High School April 26th, 2024
- Exhibitioner - Discover UH Mānoa Open House October 26th, 2024
- Exhibitioner - SOEST Open House 2023
- Participant - Argovis Hackathon 2023
- Volunteer - Orca Bowl 2020
- Volunteer - UW Society for Ecological Restoration

SKILLS & INTERESTS

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Operating System Proficiency: Windows, Mac, Linux

Experience in Programming Languages: Python, Java, MATLAB, SQL

Package Proficiency: Xarray, Pandas, netCDF4, Matplotlib, Seaborn, Seawater, Cartopy, GSW (TEOS-10), Dash, Plotly

Tools: Anaconda, Visual Studio Code, Inventor, RobotCS, Revit, Arduino, Jupyter Notebook, Microsoft Office Suite, Google Suite, ArcGIS, jGRASP, Adobe Illustrator

Geographic Information Systems: Spatial Analysis Tools, Vector and Raster Analysis, Model Builder, Surface Models, Georeferencing, Watershed Delineation

Skills: Leadership, Communication, Working in Teams, Hardware Assembly and Maintenance

Interests: Backpacking/Hiking, Cooking, Alternative/Indie Music, Traveling, Computer Building